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New palaeomagnetic data from the Mahabaleshwar Plateau, Deccan Flood Basalt Province, India: implications for the volcanostratigraphic architecture of continental flood basalt provinces

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Abstract: New magnetostratigraphic data from seven Western Ghats sections in the Deccan Volcanic Province are presented. These are combined with an established geochemically defined stratigraphy, and volcanological logs, to provide a correlated, chronological eruptive framework. We identify two magnetic polarities in five of the sections, and these are assigned to chrons 29r and 29n. Importantly, the reversal boundary represents an identifiable isochronous surface within the volcanic pile. This surface occurs at different elevations, as does the altitude of the geochemically defined formation boundary (i.e. Ambenali–Mahabaleshwar Fms), which defines a second isochronous surface. Inspection reveals significant differences in the number and thickness of lava units preserved between these two surfaces. This indicates that there was significant local topography (*c*. 80 m) across Deccan Volcanic Province lava fields during their development; an interpretation consistent with topographies observed across modern and historical examples (e.g. Hawaii, Iceland). These data also indicate that the geochemical stratigraphies of continental flood basalt provinces can mask local and sub-regional detail in lava stacking patterns when applied at smaller spatial scales ($<10^2$–10^4 m). Finally, transitional (R–N) directions preserved in some eruptive units place constraints upon the rates of eruption, and indicate *c*. 2 ka periods of volcanic repose.

Continental flood basalt provinces are large outpourings of mainly tholeiitic basalt. They are characterized by their large volume ($>1 \times 10^6$ km^3 of lava), erupted in a relatively short period of geological time (1–2 Ma: e.g. Eldholm & Coffin 2000; Jerram & Widdowson 2005). Their origin, and the possible environmental impact of the eruptions, has proven controversial, and continental flood basalt province volcanism also remains central to the continuing debate on the existence of mantle plumes, and their possible temporal correlations with Phanerozoic mass extinctions (e.g. Courtillot 1994; Courtillot *et al*. 2003; Saunders *et al*. 2007).

The link between continental flood basalt provinces and mass extinctions is thought to be due to the large outpourings of SO_2 that is degassed from the lava during eruptions (Self *et al*. 2005, 2006, 2008; Chenet 2006; Chenet *et al*. 2007, 2008), because if these SO_2 emissions rise to stratospheric altitudes and are converted to sulphate aerosols, they could have a major effect on global climate (e.g. Thordarson & Self 1996). To better understand the environmental effects of a continental flood basalt province it is important to determine the nature and architecture of the lava successions, as well as the magnitude of each eruptive event. One of the major questions regarding continental flood basalt province architecture is the style of emplacement of the lava units and, to understand this, it is crucial to study in three dimensions the meso-scale (10^{-1}–10^2 m) and macro-scale (10^2–10^4 m) features associated with the volcanism.

In this paper we present preliminary work on the lava architecture in a key part of the Western Ghats of the Deccan Trap continental flood basalt province, India, using integrated field observations, magnetostratigraphy and chemostratigraphy. The Deccan Traps are, perhaps, one of the best studied examples of a continental flood basalt province. They were erupted between 67 and 64 Ma, and thus straddle the Cretaceous–Tertiary boundary (e.g. Courtillot *et al*. 1986, 1988; Duncan & Pyle 1988; Chenet *et al*. 2007). Approximately 500 000 km^3 of lava are currently preserved (Mitchell & Widdowson 1991), but recent estimates suggest that up to 1.3×10^6 km^3 of lava may have originally been emplaced (Jay & Widdowson 2008). The bulk of these lavas are thought to have been erupted in <1 Ma (e.g. Courtillot 1988; Duncan & Pyle 1988; Chenet *et al*. 2007), and the eruption of the Deccan Volcanic Province lavas has been considered as both the prime cause of, and a contributory factor to, the mass extinctions that occurred at the Cretaceous–Tertiary boundary (e.g. Courtillot 1994; Glasby & Kunzendorf 1996; Wignall 2001; Keller *et al*. 2008).

The large-scale structure of the traps is now reasonably established, with published magnetostratigraphic and chemostratigraphic data both indicating that the traps form a broad southerly plunging anticline as a result of post-eruptive 'warping' (e.g. Wensink & Klootwijk 1971; Beane *et al*. 1986; Devey & Lightfoot 1986; Mitchell & Widdowson 1991; Vandamme & Courtillot 1992; Widdowson & Cox 1996). However, reconstructing the syneruptive architecture of the traps has proven difficult because the published palaeomagnetic, chemostratigraphic and lithostratigraphic data have not been collected simultaneously and, therefore, are difficult to place within a robust stratigraphic and geographical framework. This is because these various studies were carried out with different aims and often followed

different sampling methods, and because sampling sites may be difficult to locate given the limited precision of published maps and historical lack of the global positioning system (GPS). It is only recently with the advent of affordable GPS receivers that locating sampling sites within the Deccan Province has become quick and accurate.

Nomenclature

The complicated nature of continental flood basalt province volcanic architectures has been recognized and discussed elsewhere (e.g. Jerram 2002; Single & Jerram 2004; Jerram & Widdowson 2005). To evaluate the volcanostratigraphic architecture observed in the Western Ghats sections, three hierarchical categories (flow field, lava flow and flow lobe), devised and defined by Self *et al*. (1997), are used in this study, and are summarized here. A flow field is the aggregate product from a single eruption or vent, and is built up of one or more lava flows. For the present study, we have used flow field to describe the unit or units that occur between two weathering horizons, sometimes marked by red soils (i.e. 'boles'); this then assumes that sufficient time would have elapsed between successive eruptions at a given locality to permit at least minor weathering to occur. A lava flow is defined (Bates & Jackson 1987) as the product of a single outpouring of lava. Different lava flows are, therefore, typically formed after a short pause in effusion; however, it is possible for an eruption to form two or more lava flows simultaneously. The term flow lobe is used to describe a single part of a lava flow formed by internal inflation and that is bounded top and bottom by a glassy rind or selvage. In continental flood basalt provinces flow lobes can vary in size from 10^{-1} m (also called toes) to 10^{3} m in width and from 10^{-1} to 100 m in thickness. If a single flow lobe is a large-scale feature, and wider than an outcrop (hundreds to thousands of metres), as is common in continental flood basalt provinces, then the term sheet flow lobe (shortened to sheet lobe) is used. The term unit is used in a generic sense to describe a body of lava without any connotations as to its hierarchy within this system, or its origin or character.

Geology of the Mahabaleshwar Plateau region

The Mahabaleshwar Plateau covers an elevated region of *c*. 500 km^2 and includes a key type section of the main Deccan Province (i.e. Ambenali Ghat; Fig. 1a; Beane *et al* 1986). The plateau is situated on the edge of the immense Western Ghats Escarpment, which, on its western side, consists of a near-vertical 1400 m drop to the Konkan coastal plain (20–200 m above sea level (asl)), whereas on its eastern, inland, side there is a 500–700 m descent to the central Maharashtra Plateau (*c*. 600–800 m asl). The basalt succession that forms the Mahabaleshwar Plateau has been deformed into a gentle anticline–monocline that plunges at up to 0.3° towards the south (Devey & Lightfoot 1986; Mitchell & Widdowson 1991). Blasted road sections afford reasonable exposures through the volcanic pile, including flow contacts, which are often otherwise masked in natural outcrops. The Mahabaleshwar Plateau thus consists of an elevated area bounded on all sides by steep road ascents known locally as 'ghats', and these provide an ideal location for investigating the macro-scale 3D geometry of a key part of the Deccan lava succession. Previous workers have described this part of the Deccan Volcanic Province and, by analogy, a large proportion of continental flood basalt provinces world-wide, as having a 'layer-cake' structure because, on first inspection, it appears to consist of one extensive, flat-lying 'simple' flow on top of another.

Our work in the Deccan Volcanic Province shows that the volcanic succession is largely composed of a series of inflated pahoehoe sheet lobes (Jay 2005). In the Mahabaleshwar area, the majority of these sheet lobes are of the order of 20 ± 5 m in thickness, but there are also smaller precursor and break-out lobes, which are decimetres to metres in thickness. The latter are described as compound flows using the classification of Walker (1972).

The source vent area(s) for the lavas we investigated is not known. Locating source vents in many continental flood basalt provinces is often problematic, and this might be a function of the relative scarcity of feeder dykes, the depth of erosion (more feeders may be revealed with increasing erosion of the flood basalt stratigraphy), or, in the case of the Deccan Volcanic Province, related to rifting to the west of the study area. One of the few flood basalt lava flows that has an identifiable, associated vent region is the Roza flow in the Columbia River Basalt Province (e.g. Thordarson & Self 1998, and references therein). The source vents associated with the Roza flow manifest themselves as a series of spatter cones aligned along possible fissures (Thordarson & Self 1998). By analogy, it is assumed that the flow fields in the Deccan also issued from similar vent cones located along fissures. Given that major eruptive vents have not yet been identified in the Deccan Volcanic Province, it has been suggested that the original vent systems may lie off the western coast of India, possibly offshore of Mumbai (*c*. 18–20°N), or else have since rifted away (Mahoney 1988; Hooper 1990; Saunders *et al*. 2007).

Stratigraphy and sampling

Detailed volcanological observations combined with geochemical and palaeomagnetic sampling provide the basis for this work. As a result of the regional structure and associated southerly dip of the basalt stratigraphy (Watts & Cox 1989; Mitchell & Widdowson 1991), progressively lower stratigraphical units become exposed in the more northerly ghat sections (Fig. 1c). We studied the lava pile along the four roads that ascend the Mahabaleshwar Plateau (Ambenali Ghat; Wai–Panchgani; Kelgar road and Tapola road), together with an additional section through a stratigraphically similar part of the lava succession, 60 km to the south, along Khumbarli Ghat between Chiplun and Helwak (Fig. 1a). Two further sections were sampled to the north. Varandah Ghat, some 25 km north, is slightly lower in the stratigraphy but contains lavas of the same formations as the Mahabaleshwar Plateau traverses, with the exception of the Mahabaleshwar Formation (Fig. 1c). Matheran Ghat, 100 km to the north, is much lower in the stratigraphy and does not overlap stratigraphically with the succession of the Mahabaleshwar Plateau region.

Each section was logged in detail (Jay 2005), and collated as composite sections along the roads that ascend the spurs of the Mahabaleshwar Plateau and Western Ghats. The longest traverse (36 km long) is on Ambenali Ghat, between the towns of Poladpur and Mahabaleshwar, and is a straight-line horizontal distance of *c*. 10 km over a vertical distance of 1235 m through part of the lava pile. All the sampled sections converge on the town of Mahabaleshwar; therefore the tops of the single traverses, and their associated logs, are geographically closer together than the lowermost parts of each traverse. Detailed volcanological descriptions were made, and samples collected, of each exposed lava unit, and of any boles or weathering breaks at sheet lobe contacts along the seven road sections. For each

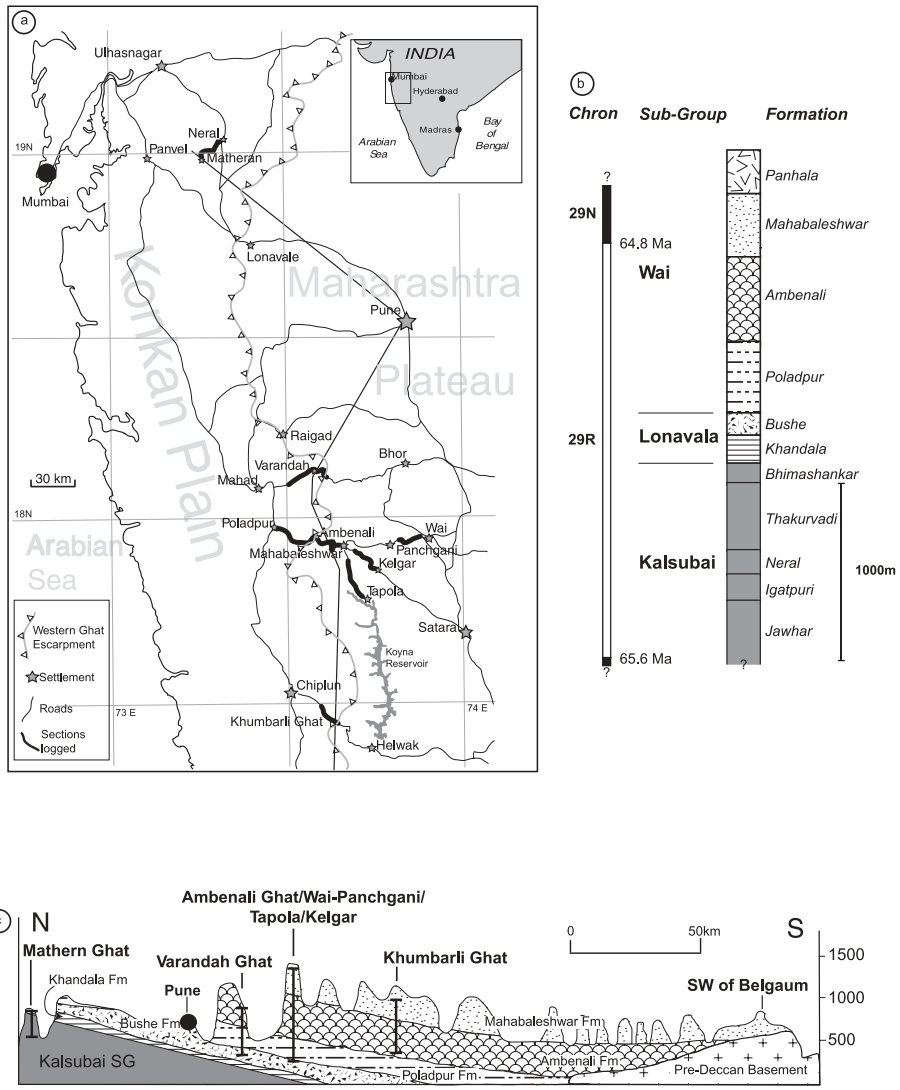


Fig. 1. (**a**) Simplified map of the Western Ghats region, indicating the location of the sampled sections and the line of cross-section. (**b**) Stratigraphic section through the Deccan Traps with formation names and relative thicknesses (after Jay & Widdowson 2008), showing the geomagnetic time scale of Cande & Kent (1995). (**c**) Schematic north–south cross-section through the Deccan traps showing the approximate location of the sampled sections; Panhala Fm and overlying Desur unit are not shown (after Beane *et al*. 1986; Devey & Lightfoot 1986; Subbarao *et al*. 1994).

sample locality, position and elevation were recorded using a hand-held GPS receiver and then cross referenced with barometric altimeters and existing topographic data. In addition, a number of reference sites were selected on the Mahabaleshwar Plateau transects and, using high-precision static GPS receivers, their locations were recorded to an accuracy of ±1.0 cm (Jay 2005). A Leica System 530 real-time GPS receiver (with the radio transmitter removed to disable the real-time capability) was positioned as a fixed base station at the Dina Hotel, Mahabaleshwar, and a second unit, the rover (also with its real-time capability disabled), was positioned at the key locations and horizons within the transects. The processing was carried out each evening, to allow a site to be revisited if necessary. These precisely positioned sites provided reference tie-points against which future barometric altimeter readings could be corrected (Jay 2005). This technique provides the most precise elevation framework yet achieved for a sampling programme in the Western Ghats region.

Samples for palaeomagnetic and geochemical analyses were collected using a petrol-powered rock drill. Cores were oriented using both magnetic and sun compasses and clinometer. Samples were collected from every exposed flow unit along each traverse as follows: 36 sheet lobes were sampled along the Ambenali Ghat traverse; 17 sheet lobes were sampled along the Wai–Panchgani traverse; 19 sheet lobes (17 for palaeomagnetic analysis) were sampled along the Kelgar road traverse; 18 sheet lobes were sampled along the Tapola traverse; 16 sheet lobes (15 for palaeomagnetic analysis) were sampled along the Khumbarli Ghat traverse; 19 sheet lobes were sampled along the Matheran Ghat traverse; 17 sheet lobes were sampled along Varandah Ghat traverse. At each locality at least two 25 mm diameter, oriented cores, were taken in the freshest available rock. At each site sister-samples were also collected from the same outcrop for geochemical analysis. The palaeomagnetic and geochemistry samples, together with the resulting analytical data, were then positioned within detailed volcanological logs (Jay 2005).

Palaeomagnetic results

Samples for palaeomagnetic analysis were cut into 22 mm long specimens in the laboratory, with each sample usually yielding at least two specimens. Four samples from the Ambenali Ghat section, and three from the Kelgar section disintegrated during sample preparation, and therefore the directions for four flows in the Ambenali section and three flows in the Kelgar section are based on demagnetization of only one sample, albeit comprising several specimens. Measurements of the natural remanent magnetization (NRM) were made using either a three-axis 2G cryogenic

magnetometer or a Geofysika Brno JR5A spinner magnetometer at the University of Oxford palaeomagnetic laboratory. Specimens were stepwise demagnetized using progressive thermal or alternating field (a.f.) techniques through at least 12 steps until they became unstable, or to 100 mT or 700 °C, respectively. Where pilot samples indicated that the NRM consisted of multiple components, the number of demagnetization steps was increased, in some cases up to 25 steps. Given that we wished to retain the samples for later geochemical analysis, the bulk of the demagnetization was carried out with the a.f. technique to avoid possible alteration associated with heating. Demagnetization results were visually inspected in orthogonal and stereographic projections. Remanence directions were determined from stable end-points using principal component analysis (Kirschvink 1980). Magnetic components were considered stable where they were defined by at least three points on vector end-point diagrams and had a maximum angular deviation (MAD) not exceeding 15°. The vast majority of samples yielding a linear component had MAD values less than 5°.

NRM intensities varied from 0.2 to *c.* 10 A m^{-1}, with a mean intensity of 2.7 A m^{-1}. The response of the samples to progressive demagnetization proved to be straightforward. After removal of a small viscous component of magnetization, typically by a.f. fields of 15 mT or temperatures of 300–350 °C, virtually all samples revealed a characteristic component that trended to the origin on orthogonal plots (Fig. 2). This component was typically removed by temperatures of 590 °C and a.f. fields of 60–90 mT, although some samples required a.f. fields of up to 140 mT to completely demagnetize them. These observations indicate that the bulk of the remanence is probably carried by magnetite (or low-Ti titanomagnetite), and are in agreement with the results of isothermal remanence acquisition experiments, where the samples typically reach saturation by direct fields of some 250 mT. The characteristic component was

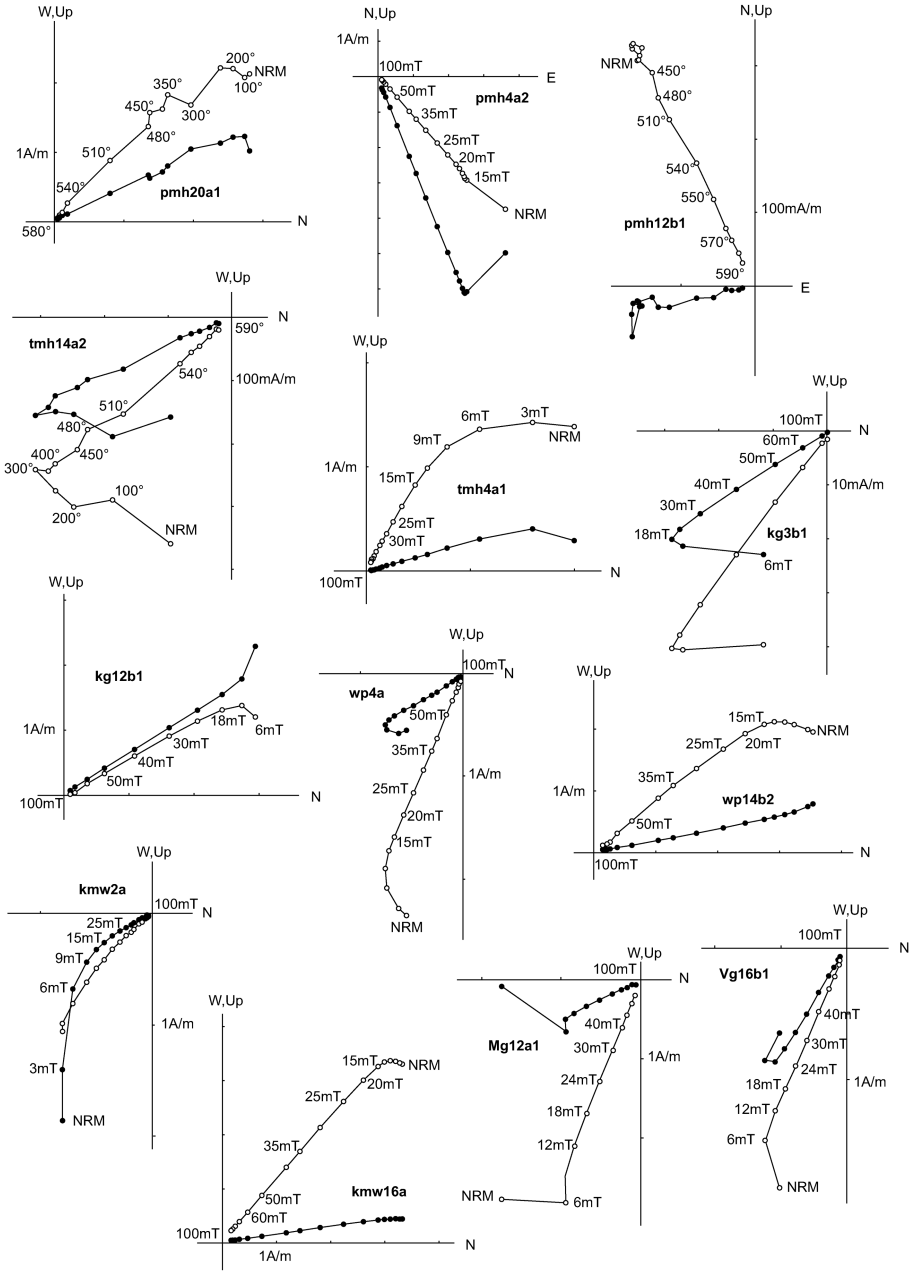


Fig. 2. Examples of demagnetization behaviour observed in each of the seven sections. An example of a normal polarity sample and reverse polarity sample is given for Ambenali Ghat (PMh20a1 and PMh4a2), Tapola (TMh14a2 and TMh4a1), Khumbarli Ghat (KG3b1 and KG12b1), Wai–Panchgani (WP4a and WP14b2) and Kelgar (KMw2a and KMw16a). Reverse polarity only was found in the Matheran Ghat (MG12a1) and Varandah Ghat (VG16b1) sections. Sample PMh12b1 exhibits a transitional direction at the 29n–29r reversal boundary in the Ambenali Ghat section. On orthogonal plots, filled (open) symbols are horizontal (vertical) projections. All plots are shown in geographical coordinates.

directed either down to the SSE, with moderate inclinations, or up to the NNW with moderate inclinations (Figs 2 and 3; Table 1). This indicates that the rocks record one reversal of the Earth's magnetic field, and this reversal was identified in five of the seven studied sections. However, because of an area of no exposure on the Tapola traverse, the elevation of the reversal horizon can be determined only to within 48 m, whereas in the other sections it is located to within a few metres.

As indicated by previous workers (e.g. Courtillot *et al*. 1986; Vandamme *et al*. 1991; Vandamme & Courtillot 1992), available palaeontology (Prasad & Khajuria 1995), and geochronology (e.g. Hofmann *et al*. 2000; Chenet *et al*. 2007), the likely chrons are 29r and 29n, the boundary between which occurs at 64.745 Ma (Cande & Kent 1995). It is noteworthy that in four of the five sections containing the reversal, the polarity change coincides with the presence of a red bole horizon, suggesting that there may have been a hiatus in volcanic activity of unknown duration at the reversal boundary (Fig. 4) over much of the Mahabaleshwar Plateau region. Also of importance is that four of the sections (i.e. Ambenali Ghat, Tapola–Mahabaleshwar, Kelgar–Mahabaleshwar and Wai–Panchgani road sections) contain basalts with transitional palaeomagnetic directions (Figs 3 and 5) suggesting that the lavas were erupted during the period while the field was reversing. However, in the Tapola and Kelgar sections the transitional directions could equally be assigned to one or other polarity; normal in the case of the Kelgar section and reverse in the case of the Tapola section.

Geochemistry

Trace element concentrations and ratios permit the sampled flow units to be assigned to formations (Jay 2005) following the Deccan Volcanic Province stratigraphy proposed by Cox & Hawkesworth (1985), Beane *et al*. (1986), Devey & Lightfoot (1986) and Mitchell & Widdowson (1991). Those workers proposed three geochemically defined subgroups (Kalsubai, Lonavala and Wai), and further subdivided them into 11 widely recognized formations (Beane *et al*. 1986; Devey & Lightfoot 1986: Fig. 1; Table 2).

The lava sequences examined in this study (excluding Matheran Ghat and the Bushe Fm in lower Varandah Ghat) are, from oldest to youngest, the Bushe, Poladpur, Ambenali and Mahabaleshwar Formations. The latter three form most of the Wai Subgroup, which is the youngest division of the Deccan Volcanic Province and is thought to be the most voluminous subgroup of lavas (Mitchell & Widdowson 1991; Widdowson *et al*. 2000). The youngest formations of the Wai Subgroup were not encountered and are not discussed further (Fig. 1c). Because there is a significant jump in the $^{87}Sr/^{86}Sr$ ratio at each formation boundary, caused by varying residence times and degrees of crustal interaction, each formation is thought to represent the outpourings of a genetically distinct, magma batch; importantly, trace element concentrations and ratios (i.e. Sr, Ba, Ba/Y and Zr/Nb) follow a similar pattern of behaviour (Mahoney *et al*. 1982; Cox & Hawkesworth 1985; Beane *et al*. 1986; Devey & Lightfoot 1986). Here, we use the trace element criteria devised by Devey & Lightfoot (1986) to identify the formations to which the sampled lava units belong (Table 2). This approach is not as discriminating as $^{87}Sr/^{86}Sr$ isotopic data, but is more readily acquired for large numbers of samples using X-ray fluorescence analysis. The lack of $^{87}Sr/^{86}Sr$ isotope data is not critical to this study as Zr/Nb ratios are known to be excellent for discriminating samples from the Ambenali and Mahabaleshwar Formations (Devey & Lightfoot 1986), and maximum and minimum values of the Zr/Nb ratios for the two formations have been collated by Widdowson *et al*. (2000). To ensure our Zr/Nb-defined

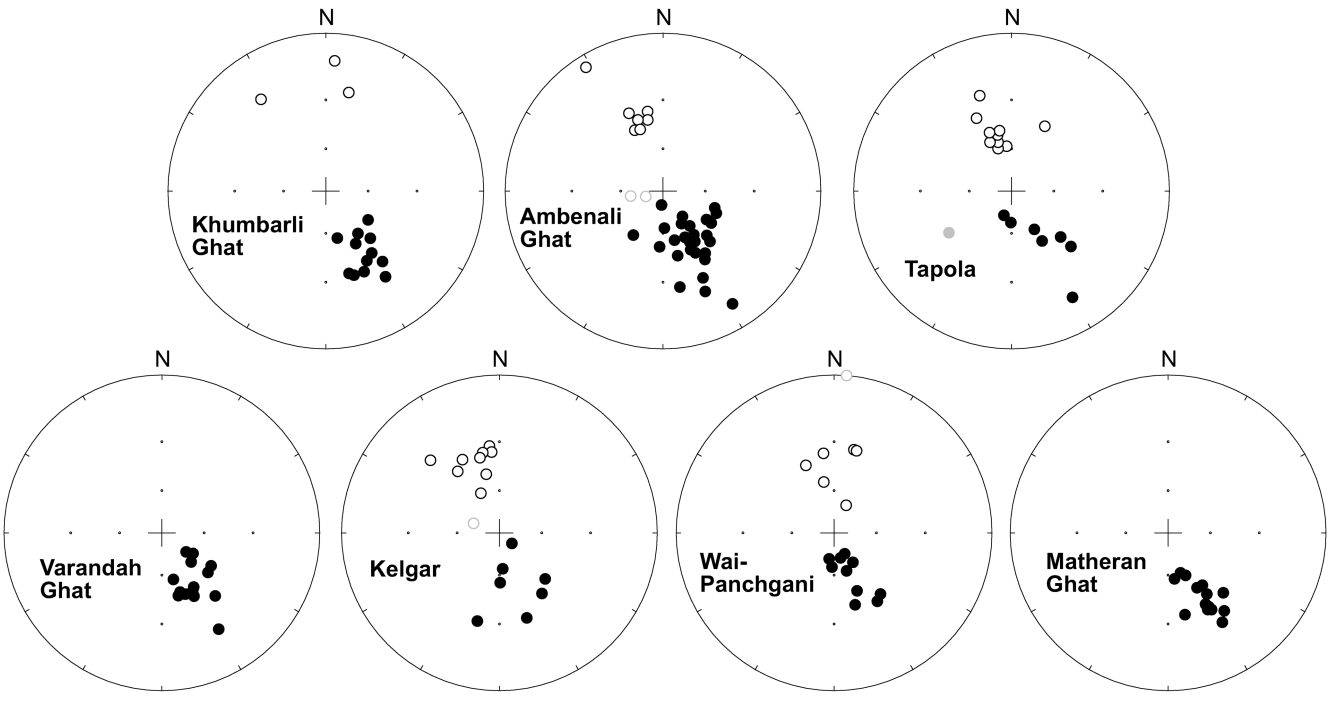


Fig. 3. Equal area projections of flow mean directions from each of studied sections. Filled (open) symbols represent lower (upper) hemisphere projections. Flows outlined in grey are samples that yielded possible transitional directions at the 29n–29r reversal boundary but have not been included in the calculation of the overall statistics.

Table 1. *Palaeomagnetic data from the Mahabeleshwar Plateau*

	D	*I*	*N*/*N*o	*R*	*k*	α_{95}	P_{lat}	P_{long}	d*p*	d*m*
Ambenali Ghat (17°56'N, 73°37'E)										
Normal (samples)	339.0	−36.2	16/16	15.49	29.4	6.9				
Reverse (samples)	152.5	47.3	47/50	43.35	12.6	6.1				
Combined (samples)	154.5	44.4	63/66	58.59	14.1	4.9	−39.5	103.7	3.9	6.2
Normal (flows)	338.9	−35.4	7	6.79	29.1	11.4				
Reverse (flows)	151.5	49.5	27	25.45	16.7	7.0				
Combined (flows)	153.5	44.7	34*		15.2	6.5	−38.8	104.5	5.2	8.2
Reversal test R^- ($\gamma_o = 15.0$, $\gamma_c = 14.58$)										
Tapola (17°48'N, 73°42'E)										
Normal (samples)	344.5	−47.4	20/20	19.28	26.5	6.5				
Reverse (samples)	151.8	52.1	15/15	13.72	10.9	12.1				
Combined (samples)	159.5	49.5	35	32.89	16.1	6.2	−38.0	096.2	5.5	8.2
Normal (flows)	346.6	−48.4	10	9.7	29.7	9.0				
Reverse (flows)	150.2	50.4	7	6.51	12.2	18.0				
Combined (flows)	160.2	49.5	17*		18.5	8.3	−38.2	095.5	7.3	11.0
Reversal test R^+ (c) ($\gamma_o = 10.84$, $\gamma_c = 17.29$)										
Kelgar (17°52'N, 73°44'E)										
Normal (samples)	341.6	−40.5	18/19	17.55	38.0	5.7				
Reverse (samples)	168.2	53.6	11/11	10.02	10.2	15.0				
Combined (samples)	163.6	45.3	29/30	27.38	17.3	6.6	−42.6	093.7	5.3	8.4
Normal (flows)	339.7	−40.0	9	8.75	32.3	9.2				
Reverse (flows)	163.0	50.8	7	6.53	12.7	17.6				
Combined (flows)	161.3	44.6	16*		19.1	8.7	−42.3	096.6	6.9	11.0
Reversal test R^+ (c) ($\gamma_o = 11.04$, $\gamma_c = 17.45$)										
Wai–Panchgani (17°56'N, 73°50'E)										
Normal (samples)	359.0	−45.0	13/13	12.43	21.2	9.2				
Reverse (samples)	159.5	57.5	20/20	19.21	24.0	6.8				
Combined (samples)	168.4	53.0	33	31.3	18.8	5.9	−37.3	086.0	5.7	8.2
Normal (flows)	359.7	−45.7	6	5.74	18.9	15.8				
Reverse (flows)	159.5	57.5	10	9.62	23.54	10.2				
Combined (flows)	168.3	53.5	16*		18.6	8.8	−36.8	085.9	8.5	12.3
Reversal test R^- ($\gamma_o = 17.1$, $\gamma_c = 16.86$)										
Khumbarli Ghat (17°25'N, 73°40'E)										
Normal (samples)	350.2	−18.3	5/6	4.67	12.0	23.0				
Reverse (samples)	149.7	40.8	23/24	22.25	29.2	5.7				
Combined (samples)	153.9	37.3	28/30	26.44	17.3	6.7	−44.0	108.5	4.6	7.9
Normal (flows)	354.0	−19.8	3	2.81	10.6	39.9				
Reverse (flows)	149.3	40.4	12	11.67	33.8	7.6				
Combined (flows)	154.8	36.9	15/15		17.2	9.5	−44.7	107.8	6.5	11.1
Reversal test R^o ($\gamma_o = 29.48$, $\gamma_c = 40.21$)										
Varandah Ghat (18°08'N, 73°35'E)										
Normal (samples)	149	48.3	30/35		28.9	5.1	−33.9	106.4	4.4	6.7
Normal (flows)	148.5	48.5	15/17		30.0	7.2	−33.5	106.7	6.2	9.5
Matheran Ghat (19°01'N, 73°18'E)										
Normal (samples)	149.3	43.3	32/38		29.9	4.7	−36.6	108.4	3.6	5.8
Normal (flows)	149.3	43.3	17/19		31	6.5	−36.6	108.4	5.0	8.1

Latitude and longitude of the sampled section are given. *D* and *I*, mean declination and inclination (based on samples or flows) of the section; *N*/*N*o, number of samples (or flows) used in the calculation of the mean direction/number of samples (or flows) demagnetized; *k*, Fisher (1953) precision parameter; α_{95}, half-angle of the cone of 95% confidence about the mean direction; P_{lat} and P_{long} latitude and longitude of the resultant palaeopole for each section; d*p* and d*m*, semi-axis of 95% confidence about the palaeopole. The reversal test used is that of McFadden & McElhinny (1990): R^+, positive result; R^-, negative result; R^o, indeterminate result. γ_o and γ_c, observed and critical values of gamma, respectively.

*For the Ambenali section two of the 36 flows sampled yielded transitional directions; for the Tapola, Kelgar and Wai–Panchgani sections one flow in each section yielded transitional directions. These have not been included in the statistics in this table.

Ambenali–Mahabaleshwar formation boundary is compatible with the original, $^{87}Sr/^{86}Sr$-defined, boundary (Devey 1986), we have accurately determined its position on the new Ambenali, Kelgar and Wai–Panchgani section logs. Our Zr/Nb-defined formation boundary is within *c*. 10 m (one sheet lobe) of the $^{87}Sr/^{86}Sr$-defined boundary on the Ambenali and Kelgar sections, and on the Wai–Panchgani section it is within *c*. 30 m (two sheet lobes) (Figs 4 and 6).

Integration of palaeomagnetic, geochemical and volcanological data

The location of the chemostratigraphic boundaries relevant to this study in the five southern sections are shown in Figure 4, together with the relevant portions of the volcanological logs. Figure 4 also depicts the palaeomagnetic reversal (29r–29n) recorded in five of the sections, and forms the basis for

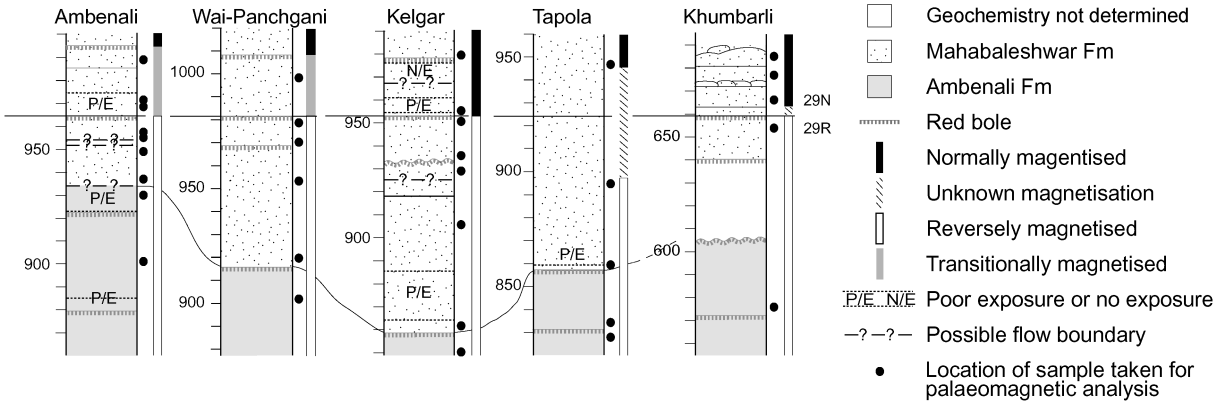


Fig. 4. Comparison of the five sections where the palaeomagnetic reversal (29r–29n) has been observed, showing the sampling locations (•). The stratigraphic columns have been aligned along the reversal boundary, and the geochemical boundary between the Ambenali and Mahabaleshwar Formations is also indicated. The locations of several bole horizons are also plotted. It can readily be observed that the geochemical Ambenali–Mahabaleshwar Formation boundary does not align with the reversal boundary and, given that the two should represent isochronous surfaces, this indicates that there was significant relative topography between the sections during the emplacement of the Deccan Volcanic Province. This is illustrated (at a very much exaggerated vertical scale) by the fine black line joining all the sections.

the following descriptions. Two of the five sections pass a reversal test, one yields an indeterminate result, and two narrowly fail. It is noteworthy, however, that the normal polarity inclinations are all shallower than their reverse counterparts and this might be due to the rapid northward drift of India into lower latitudes during the eruption of the Deccan Traps. We now outline the relationship between the reversal boundary and the top of the Ambenali Formation in each of these five sections.

Ambenali Ghat

All 36 flows analysed from the Ambenali section yielded interpretable palaeomagnetic data: 27 flows yielded samples of reverse polarity, two yielded transitional directions, and seven yielded normal polarity data. The data narrowly fail the McFadden & McElhinny (1990) reversal test, although it noteworthy that they would pass if the flow with the shallowest normal polarity inclination were excluded. We find no compelling reason, however, to exclude this data point. On Ambenali Ghat the last reverse polarity samples (29r) are just below a red bole horizon, some 30 m above the top of the Ambenali Formation. The transitional direction occurs above a bole horizon, and the first normal polarity samples (29n) are located 20 m above the bole. The distance of 30 m between the top of the Ambenali Formation and the top of Chron 29r is the smallest thickness of material between these two horizons on any traverse. This thickness represents four sheet flows and one flow field because this part of the succession is bounded by identifiable bole or weathering horizons.

Wai–Panchgani road

All 17 flows analysed from the Wai–Panchgani section yielded interpretable palaeomagnetic data: 10 flows yielded samples of reverse polarity, one yielded a transitional directions, and six yielded normal polarity data. As with the Ambenali section, the data narrowly fail the McFadden & McElhinny (1990) reversal test. In the Wai–Panchgani section a thin red bole occurs at the reversal horizon, and units above and below it consist of thick, well-preserved inflated pahoehoe sheet (*c.* 10–20 m) lobes that form prominent vertical cliffs. On this traverse the last reverse polarity sample (29r) observed is some 65 m above the top of the Ambenali Formation (i.e. within the base of the Mahabaleshwar Formation). A transitional direction sheet lobe occurs immediately above the bole, and the first definitively normal polarity samples (29n) are located 65 m above the bole (which almost certainly represents the next flow field in the succession after the transitional direction sheet lobe). Accordingly, there is 65 m of Mahabaleshwar Formation basalt that occurs between the top of the Ambenali Formation and the top of Chron 29r. This consists of two sheet flows and two flow fields.

Kelgar–Mahabaleshwar road

On the Kelgar section an equivalent red bole can be identified at the reversal horizon. Of the 17 flows analysed for palaeomagnetic data, seven yielded reverse polarity data, one yielded a possible transitional direction and nine yielded normal polarity directions. The data pass the McFadden & McElhinny (1990) reversal test. Here, the last reverse polarity sample (29r) occurs some 95 m above the top of the Ambenali Formation. One sample above this yields a possible transitional direction (Fig. 3), although this could also be taken as belonging to the normal polarity population. If it is taken to be a normal polarity, then the position of the reversal boundary may be constrained to within a metre. Inspection of the lava-flow characteristics indicates that at least three sheet lobes are present within two flow fields between the top of the Ambenali Formation and the last reverse polarity sample.

Tapola–Mahabaleshwar road

On the Tapola section, the reversal horizon occurs within an area of poor exposure. Of the 18 flows analysed, seven yielded reverse polarity directions, one yielded a possible transitional direction and 10 recorded normal polarity directions. The data pass the McFadden & McElhinny (1990) reversal test. The last undoubted reverse polarity sample (29r) is located 2 m above of the top of the Ambenali Formation, and the first undoubted normal polarity sample (29n) some 90 m above this boundary. It is unclear whether the transitional flow depicted in Figure 3 should be included in the reverse polarity group or is truly transitional. If it is taken to be a reverse polarity sample, then there is a minimum

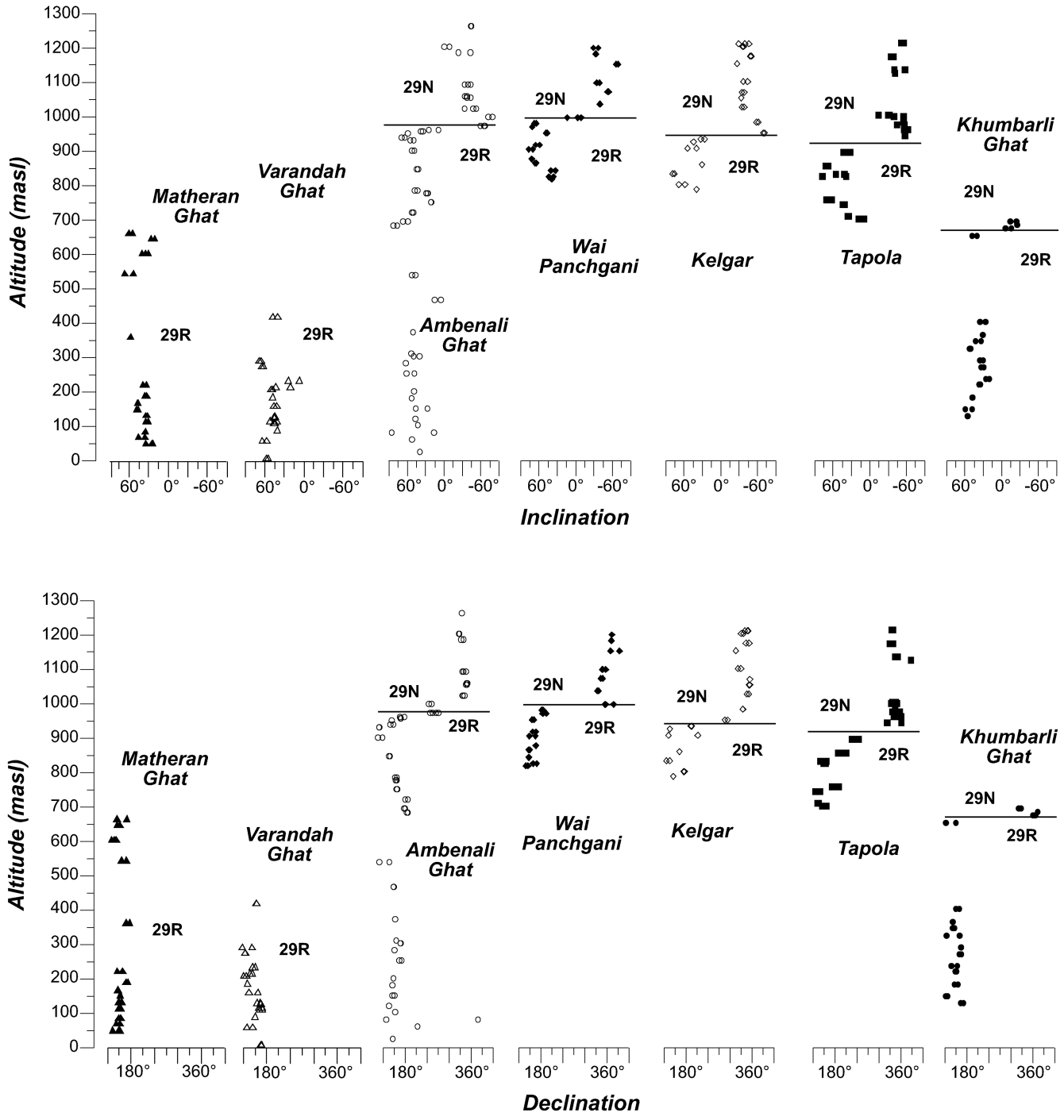


Fig. 5. Magnetostratigraphic correlation of each of the sections. It can clearly be seen that the 29n–29r reversal boundary occurs at a different altitude within each section, indicating either that there has been differential vertical movement between the various sections post-dating their emplacement or, more likely, that there was substantial topography present when the lavas were being erupted at this boundary.

of 48 m of section between the top of the Ambenali Formation and the reversal boundary.

Khumbarli Ghat

On the Khumbarli Ghat section all 15 flows analysed for palaeomagnetic data yielded clear results. Twelve flows yielded reverse polarity directions, and three flows yielded normal polarity directions. The McFadden & McElhinny (1990) reversal test yielded an indeterminate result, probably because of the small numbers of flows of normal polarity. In this section the last reverse polarity sample (29r) is identified some 55 m above the top of the Ambenali Formation. The first normal polarity sample (29n) is located 70 m above the top of the Ambenali Formation. The volcanological characteristics of the basalt successions at and above the first normal polarity sample are different from those observed on the Mahabaleshwar Plateau; here, small pahoehoe lobes with a maximum thickness of 2 m

Table 2. *Criteria used to determine the geochemical formations of the studied sections, from Devey & Lightfoot (1986)*

Formation	Criteria				
	Sr (ppm)	Ba (ppm)	Ba/Y	$^{87}Sr/^{86}Sr$	Zr/Nb
Panhala	<200	<90	–	–	>13
Mahabaleshwar	>250	>100	>4	>0.705	<10.5
Ambenali	200–250	<100	<3.5	<0.705	10.5–15
Poladpur	–	>100	>3.5	0.705–0.712	15–20
Bushe	–	>100	–	>0.713	>20

In this paper we use trace element ratios and concentrations only. Zr/Nd ratios are the most appropriate identifier of the Ambenali–Mahabaleshwar Formation boundary in the absence of $^{87}Sr/^{86}Sr$ data.

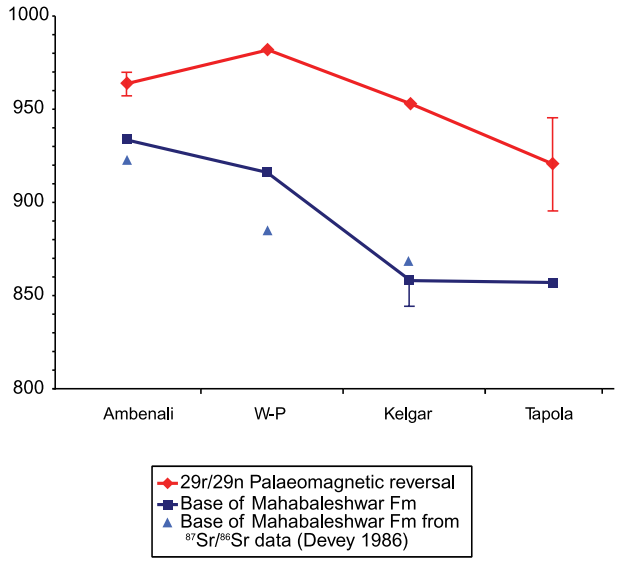


Fig. 6. Diagram showing the difference in thickness of lava between the base of the Mahabaleshwar Formation and the 29r–29n palaeomagnetic reversal horizon. Also shown is the elevation of the base of the Mahabaleshwar Formation from Cox & Hawkesworth (1985) and Devey (1986), derived from combined $^{87}Sr/^{86}Sr$ isotopic ratios, trace element ratios and concentrations. These data are not available for the Tapola road.

are common rather than the thick sheet lobes of the Mahabaleshwar sections.

Varandah Ghat and Matheran Ghat

Neither of these sections have a palaeomagnetic reversal but, as expected, the polarity is reversed (29r).

Discussion

All the exposed lava units and formations we have studied are dominated by inflated pahoehoe sheet flow lobes. The majority of these lobes are *c*. 20 (± 5) m thick (Jay 2005). Some sheet lobes have small, underlying, precursor pahoehoe lobes (up to 2 m thick and tens of metres long) representing the early stage of flow advancement; these were also thickened by internal inflation processes, and, occasionally, similar-sized break-out lobes are seen lying on top of the major sheet lobes. In some parts of the sections, especially towards the top of Khumbarli Ghat, there are stacks of small pahoehoe lobes, some as small as 10 cm across, forming a hummocky pahoehoe flow field (Jay 2005).

The boundary between the base of the Mahabaleshwar Formation and the top of the Ambenali Formation appears to be marked by a bole or weathering horizon in three of our sections (Fig. 4), and because the formation boundaries mark a change in the nature of the magma source of the chemostratigraphical lava units, this can be taken as an essentially isochronous tie-point. Importantly, the position of the polarity change from 29r to 29n occurs towards the base of the Mahabaleshwar Formation at varying heights above the lowermost geochemically defined Mahabaleshwar Formation sheet lobe. Therefore, because the geomagnetic reversal also marks an essentially isochronous tie-line between the sections, it can clearly be seen that different thicknesses of Mahabaleshwar Formation lava and different numbers of sheet lobes and flow fields are preserved between the two chronological tie-points at the different localities (Fig. 4). This difference in thickness of lava precludes post-emplacement tilting or warping of the lava succession as being the sole cause of the varying heights of either the palaeomagnetic reversal horizon or the geochemically defined boundary. Clearly, if this were the case, the two isochronous surfaces would then have a constant offset. Instead, the differences observed between the sections suggest that up to 80 m of topography was developed across the surfaces of the previously erupted units (Fig. 6).

The origin of topography is demonstrated by the different numbers of sheet flows and/or flow fields between the two isochronous surfaces in the different sections, as these indicate that across the Mahabaleshwar plateau area varying numbers of eruption products occurred between the beginning of the Mahabaleshwar Formation and the palaeomagnetic reversal. For instance, the number of flow fields varies, with one in Ambenali Ghat, but two in the other traverses (except Tapola, where the exposure is poor). With respect to sheet lobes, there are four on Ambenali Ghat, three on the Kelgar road, and only two on the Wai–Panchgani road and Khumbarli Ghat (Fig. 4; again Tapola has exposure that is poor).

This variation in the number of flow fields and sheet lobes in different sections demonstrates the limited lateral extent of the volcanological units, as they must be narrower than the width of the Mahabaleshwar Plateau (*c*. 35 km). For sheet lobes this is best shown by those with transitional palaeomagnetic directions. Two sections yield directions that are almost certainly transitional (Ambenali Ghat and Wai–Panchgani road), but these transitional directions are very different, even though they were both emplaced during the field reversal episode. They are, therefore, different sheet lobes and could even be different lava flows. The occurrence of sheet lobes with different transitional directions in different traverses is good evidence that the sheet lobes were confined, or of limited lateral extent, and did not spread across the whole Mahabaleshwar Plateau area. Further evidence can be taken from the very different morphology of the lobes above the

palaeomagnetic reversal on Khumbarli Ghat. Here, the volcanic architecture is characterized by small pahoehoe lobes and toes, rather than the thick sheet lobes observed near the reversal on the Mahabaleshwar Plateau traverses.

This variation and complexity in the lava pile is most simply explained by the emplacement of lavas forming topography on the active surface of the Deccan Volcanic Province. In most cases lava will flow towards the lowest point and so lower-lying areas should accumulate more lava than those that are elevated. Once the inflating sheet lobe fills in a depression, further inflation could cause the sheet lobe to rise above the surrounding lava and a topographic high would develop with respect to the surrounding older units (Fig. 7).

These observations all suggest that significant local topography existed in the Deccan Volcanic Province during the construction of the Traps succession. Moreover, it can also be assumed that similar relationships existed throughout the western Deccan succession and, that if topography of up to 80 m occurred across neo-formed lava surfaces during the eruptive episode, it could have locally controlled the channelling of subsequent eruptive units or else provided depressions in which shallow pools, ponds and localized sedimentation may have occurred. Furthermore, the 'layer-cake' stratigraphy often attributed to flood basalt provinces is a misnomer that serves to mask the complex interaction of erupting flow-units, pre-existing topography and localized sedimentary deposits. Accordingly, caution should be exercised in applying the broad approach offered by geochemical or palaeomagnetic stratigraphies to attempts at correlating continental flood basalt province volcanic successions at the sub-regional or local scales (i.e. 10^2–10^4 m).

The occurrence of magnetically transitional sheet lobes also gives us an insight into the eruption frequency in the Deccan Volcanic Province because a palaeomagnetic reversal is estimated to take *c*. 5–10 ka (Merrill *et al*. 1996). The single transitional sheet lobes on Ambenali Ghat (two flows) and Wai–Panchgani road (one flow) (Fig. 4) are bounded by reversely magnetized flow fields below and normally magnetized flow fields above. Therefore, the minimum period of time between the emplacement of these reverse polarity and normal polarity flow fields is 5 ka. In the case of the sections where these transitional flows occur, the minimum average eruption rate is, therefore, approximately one eruption every 2 ka (i.e. one flow field, the product of a single eruption or vent, every 2 ka). Because the transitional directions in the sheet lobes of Ambenali Ghat are different from those preserved in the Wai–Panchgani road traverse, the eruptions that produced the transitional sheet lobes must have been intermittently active during the time it took for the Earth's magnetic field to reverse. In other words, the flow field was actively being emplaced at different times and at different localities across the Mahabaleshwar area.

The implications for such intermittent activity deserve further consideration. Given that major flow fields are believed to develop over periods of 10–100 years (Self *et al*. 1997, 2006), a frequency of one eruption every 2 ka does not appear to support models (e.g. Chenet *et al*. 2008) advocating very rapid eruption of the whole Deccan succession. However, although the Mahabaleshwar area contains one of the thickest of the preserved successions, it should nevertheless be considered in the context of the wider evolution of the Deccan continental flood basalt province. It seems likely that adjacent regions and those further afield within the evolving continental flood basalt province would have experienced lava emplacement and advancement, for instance elsewhere in the flow field, during any long quiescent phases experienced in the Mahabaleshwar area. If this assumption is correct, then the construction of the Deccan continental flood basalt province was probably the result of quasi-continuous eruption, with different regions of the evolving lava fields becoming active at different times. Importantly, this type of eruptive model has the capability of generating a mosaic of highly dynamic, long-lived volcanic environments within the limits, and across the immediate vicinity of, the developing continental flood basalt province.

Conclusions

(1) The 29r–29n palaeomagnetic reversal is found at the base of the Mahabaleshwar Formation in the five most southerly traverses, and its elevation has been precisely positioned using a high-resolution GPS.

(2) The thickness of basalt between the palaeomagnetic reversal horizon and the base of the Mahabaleshwar Formation, and the associated variation in the number of preserved sheet lobes and flow fields, suggests that there was considerable topography of up to 80 m on this surface of the Deccan Volcanic Province whilst it was active.

(3) Both chemo- and palaeomagnetic stratigraphies are clearly useful at the broader scale, to elucidate the structure and post-eruptive deformation of the whole Deccan Volcanic Province. However, care must be taken when using either approach to determine stratigraphical markers, because the topography that developed on the active surface becomes important at the local and sub-regional scales.

(4) The variation in the number of sheet lobes and flow fields between the palaeomagnetic reversal and the base of the Mahabaleshwar Formation suggests that not all sheet lobes occur in each traverse and, therefore, the lateral extent of the sheet lobes and possibly flow fields in the Deccan Volcanic Province does not extend across the Mahabaleshwar Plateau. In effect, they can be continuous only over distances of less than *c*. 35 km.

(5) The current thinking that thick sheet lobes, often referred to as simple flows, are continuous over one region needs to be re-evaluated. In contrast, flow fields, comprising one or more

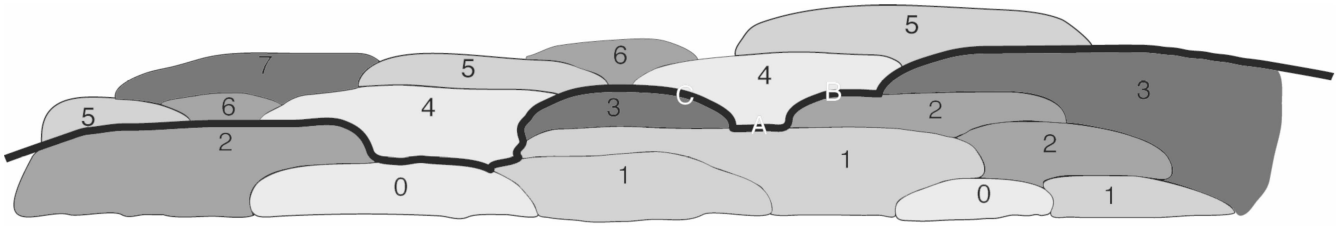


Fig. 7. A schematic illustration (not to scale) demonstrating the complexities of a section of sheet lobes in a flood basalt province. Each number indicates a different flow-field (0 is oldest, 7 is youngest). This shows that the elevation of the younger sheet lobes can be stratigraphically lower than the adjacent older units. The bold line is an isochronous horizon (e.g. the 29r–29n reversal boundary).

lava flows from a single eruptive event, may be large in total area (still as yet undetermined for the Deccan Volcanic Province), but still small in comparison with the Deccan Volcanic Province as a whole. The recognition of this nested hierarchy of scales (sheet lobe, lava flow and flow field) further demonstrates that the volcanological architecture of continental flood basalt provinces is highly complex.

(6) The term 'layer-cake stratigraphy' should be used with some circumspection where describing continental flood basalt provinces.

(7) The occurrence of transitional direction sheet lobes indicates that the minimum eruption rate at a given locality is approximately one eruption every 2 ka. Although this may appear to suggest intermittent activity interspersed with long periods of volcanic quiescence, activity in the continental flood basalt province as a whole is likely to have been quasi-continuous, as regions of evolving flow fields may have been active elsewhere.

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